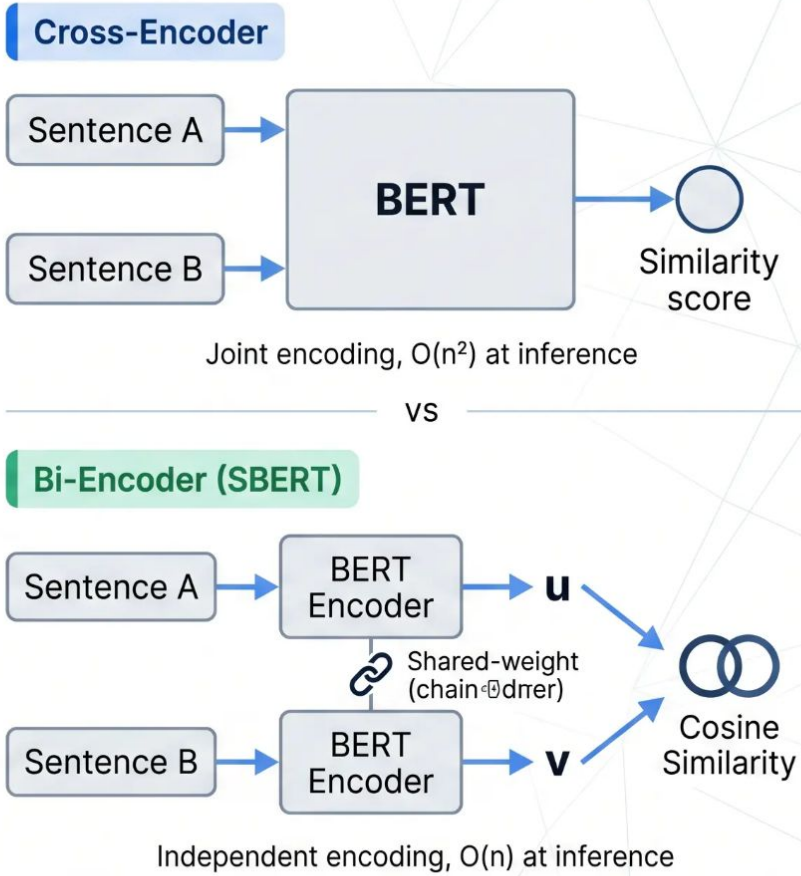
NLP · Deep Learning

Sentence BERT for Information Retrieval

Bi-Encoders and Cross-Encoders



❓ Question

> Which pair is more similar?
>
> A: “A man is eating pizza”
> B: “A person is having food”
>
> OR
>
> A: “A man is eating pizza”
> C: “A man is driving a car”

Before SBERT: Traditional Sentence Representations

Three generations of sentence encoding — and why each falls short for large-scale IR

TF-IDF & BM25

Lexical Retrieval

- Matches exact keywords — ignores word meaning
- 'bank' (finance) = 'bank' (river)
- Fast & scalable, but vocabulary-bound

✕ No Semantic Understanding

GloVe / Word2Vec

Static Word Embeddings

- Average word vectors → one sentence vector
- Captures general word semantics
- Context-free: same vector regardless of sentence meaning
- 'bank' always has the same embedding

✕ Context-Free

Vanilla BERT

Cross-Encoder

- Encodes sentence pairs jointly (full attention)
- High accuracy — but one forward pass per pair
- 10K sentences → ~50M comparisons
- ~65 hours for a 10K corpus

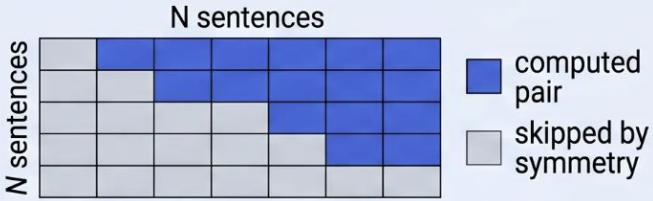
✕ $O(n^2)$ Cost

The gap: We need embeddings that are semantically rich AND fast enough to scale to millions of sentences.

Limitations of Vanilla BERT

Why vanilla BERT cannot scale for large-scale sentence comparison

The N^2 Comparison Problem



10,000 sentences → ~50 Million pairs

Each pair needs one full BERT forward pass
Result: **~65 hours** on modern GPU

- Comparing N sentences = $N \times (N-1)/2$ forward passes
- Complexity: $O(n^2)$ — quadratic growth
- Doubling corpus size → 4× the compute

✕ Infeasible for IR at Scale

Poor Sentence Embedding Quality

> [CLS] and mean-pooled BERT embeddings score BELOW GloVe on STS benchmarks

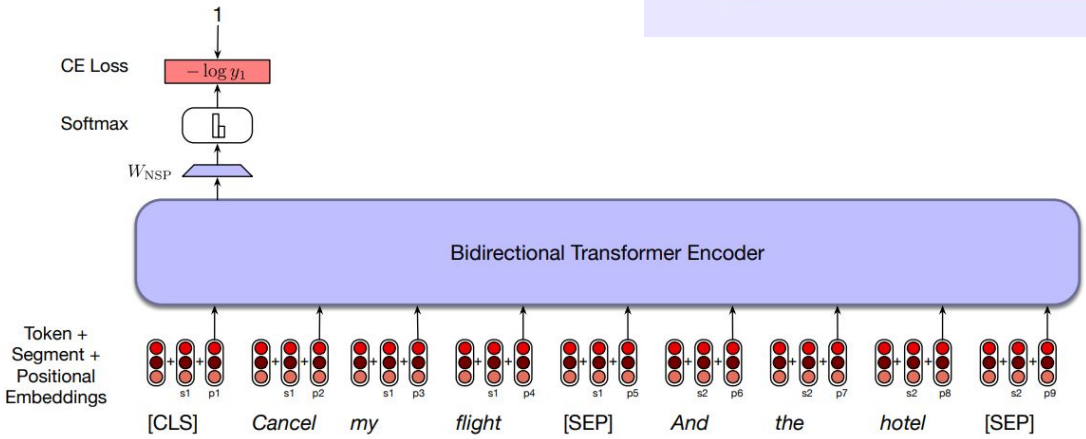
- [CLS] token not trained for sentence semantics
- Mean pooling ignores token importance
- Cosine similarity outputs are poorly calibrated

✕ Worse Than GloVe on STS

Pre-Training tasks

Next Sentence Prediction (NSP) - BERT

- Given two sentences, predict whether the second one actually follows the first one in the text.



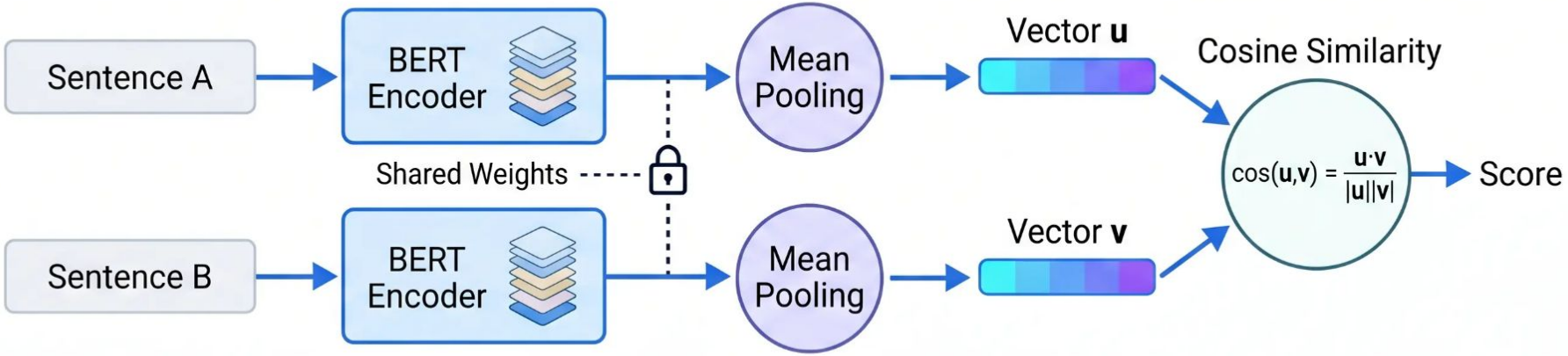
During training, the output vector from the final layer associated with the [CLS] token represents the next sentence prediction. A learned set of classification weights is used to produce a two-class prediction from the raw [CLS] vector

Model	STS12	STS13	STS14	STS15	STS16	STSb	SICK-R	Avg.
Avg. GloVe embeddings	55.14	70.66	59.73	68.25	63.66	58.02	53.76	61.32
Avg. BERT embeddings	38.78	57.98	57.98	63.15	61.06	46.35	58.40	54.81
BERT CLS-vector	20.16	30.01	20.09	36.88	38.08	16.50	42.63	29.19
InferSent - Glove	52.86	66.75	62.15	72.77	66.87	68.03	65.65	65.01
Universal Sentence Encoder	64.49	67.80	64.61	76.83	73.18	74.92	**76.69**	71.22
SBERT-NLI-base	70.97	76.53	73.19	79.09	74.30	77.03	72.91	74.89
SBERT-NLI-large	72.27	**78.46**	**74.90**	80.99	76.25	**79.23**	73.75	76.55
SRoBERTa-NLI-base	71.54	72.49	70.80	78.74	73.69	77.77	74.46	74.21
SRoBERTa-NLI-large	**74.53**	77.00	73.18	**81.85**	**76.82**	79.10	74.29	**76.68**

Table 1: Spearman rank correlation ρ between the cosine similarity of sentence representations and the gold labels for various Textual Similarity (STS) tasks. Performance is reported by convention as $\rho \times 100$. STS12-STS16: SemEval 2012-2016, STSb: STSbenchmark, SICK-R: SICK relatedness dataset.

SBERT Architecture: Siamese Networks

Shared-weight encoders with pooling to produce fixed-size sentence vectors



Shared Weights

Both encoders share identical parameters. Updating one updates both, ensuring consistency.

Mean Pooling

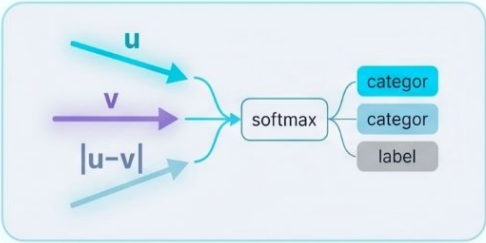
Token embeddings are averaged to produce a single fixed-size vector representation.

Cosine Similarity

Semantically similar sentences produce vectors close together in the embedding space.

Training Objectives

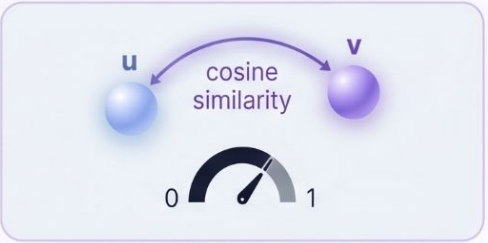
How SBERT learns meaningful sentence embeddings



Classification

Concatenates **u**, **v**, and **|u−v|** then passes through a softmax layer to predict a relationship label.

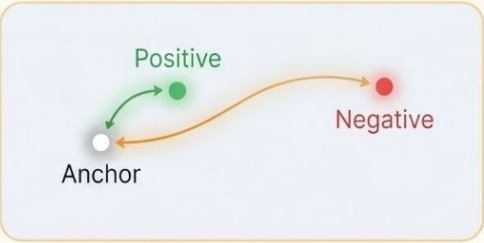
Used for: NLI tasks



Regression

Computes **cosine similarity** between embeddings **u** and **v** and minimizes the distance to a target

Used for: STS tasks



Triplet Loss

Pulls the **anchor** closer to the **positive** and pushes it away from the **negative** in embedding space.

Used for: Metric learning

"All three objectives share one goal: sentences with similar meaning should be close together in vector space."

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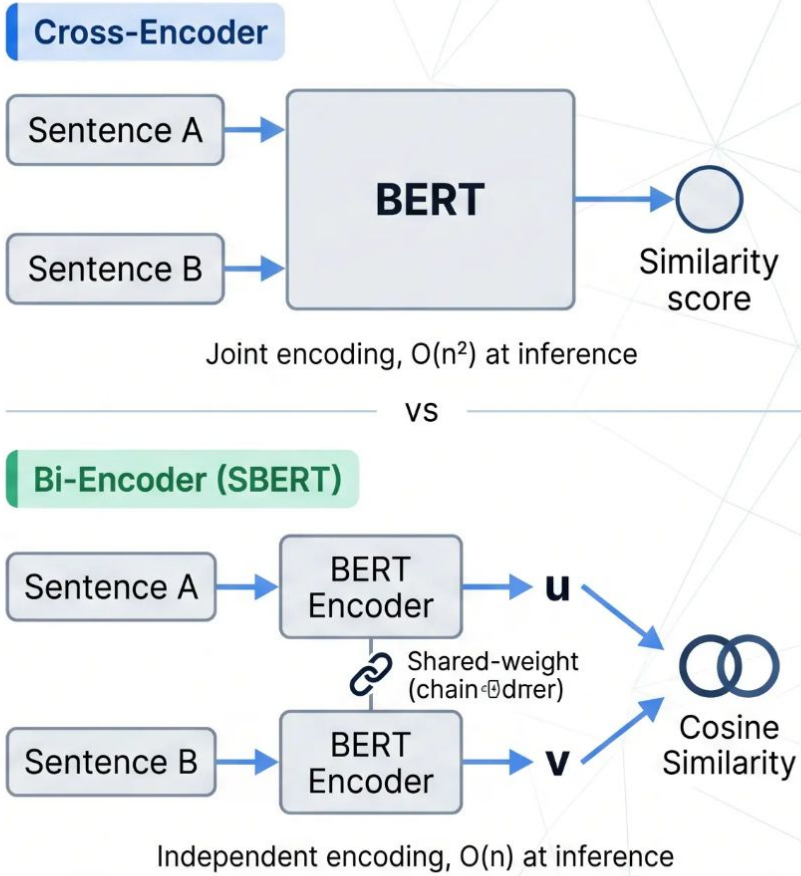
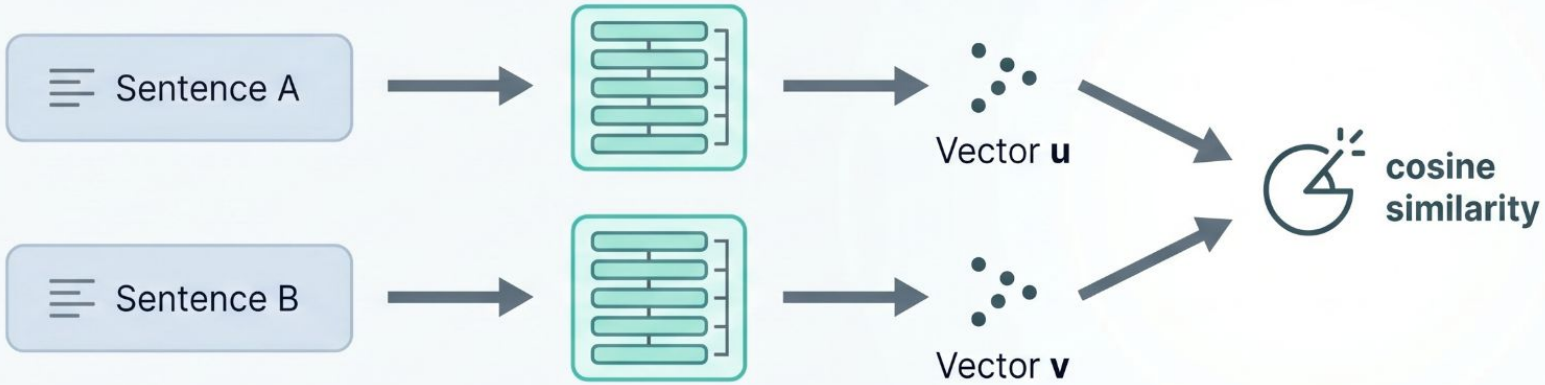
Bi-Encoders and Cross-Encoders



Bi-Encoders for Efficient Retrieval

"Encode once. Search anywhere. Instantly."



Speed

~5 Seconds

Finds most similar pair among 10,000 sentences — vs. 65 hours with vanilla BERT

Pre-computation

Encode Once

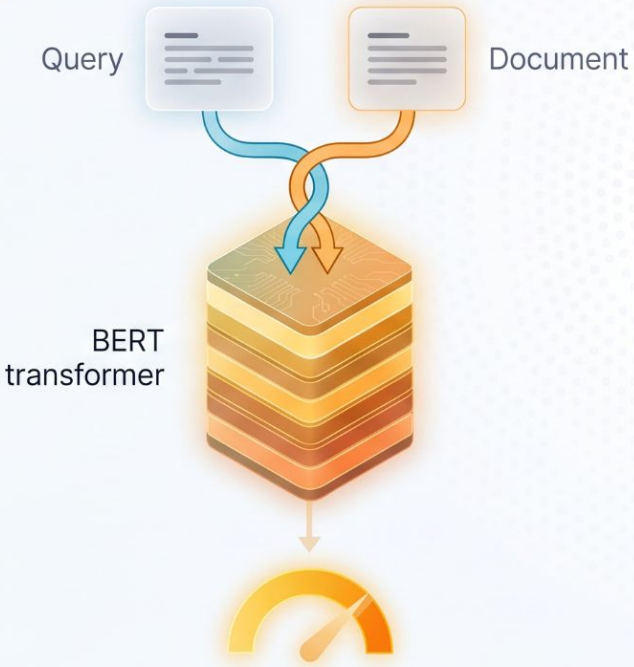
Corpus embeddings computed offline and stored — no re-encoding at query time

Scalable Index

FAISS Ready

Vectors indexed for approximate nearest-neighbor search across millions of documents

Cross-Encoders for High Precision



HOW IT WORKS

Both the query and document are passed **together** into a single BERT model, enabling every token to attend to every other token across both inputs.

STRENGTH

Captures **fine-grained interactions** between query and document tokens — understanding nuance, context, and relevance at the deepest level.

LIMITATION

Cannot pre-compute embeddings. Every new query requires **N full BERT passes** — one per document. Impractical for large-scale retrieval alone.

⚡ Bi-Encoder	🎯 **Cross-Encoder**
~5 seconds for 10K sentences	**~65 hours** for 10K sentences
Lower accuracy	**Superior accuracy**

Bi-Encoders vs. Cross-Encoders

Two fundamentally different encoding strategies

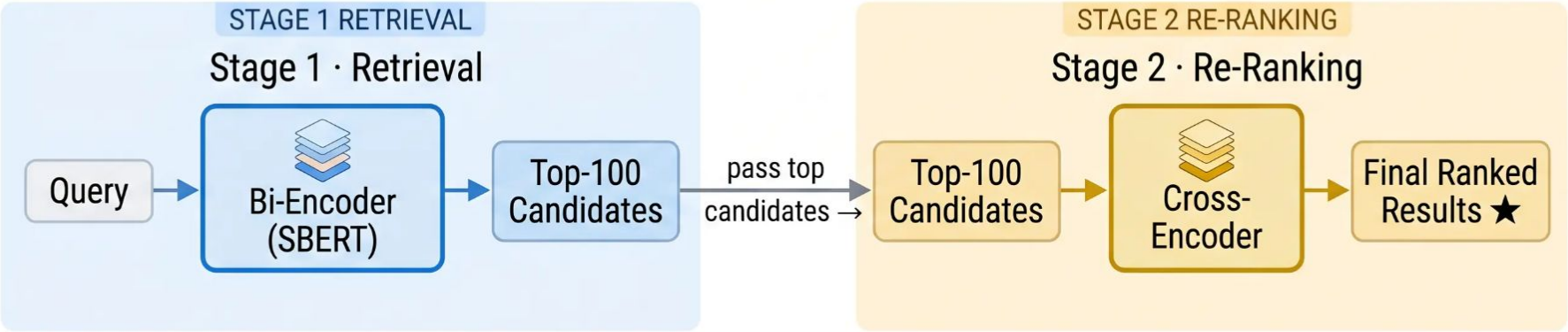
Feature	Bi-Encoder	Cross-Encoder
Architecture	Encodes each sentence independently	Encodes query + doc jointly
Speed	⚡ ~5 seconds for 10K pairs	🐢 ~65 hours for 10K pairs
Accuracy	Good — cosine similarity	Superior — full token attention
Pre-computation	✔ Embeddings stored offline	❌ Cannot pre-compute
Best Use	Large-scale retrieval	Re-ranking top candidates

💬 Discussion Question

Given the trade-offs, when would you choose a Cross-Encoder over a Bi-Encoder — and why?

Retrieve & Re-Rank Pipeline

Combining speed and precision in two stages



⚡ ~Milliseconds · Pre-computed embeddings

🎯 High precision · Full attention over query & docs

💬 Discussion Question

Why is it impractical to apply the Cross-Encoder directly to the entire corpus instead of just the top candidates?

❓ Question

> You have 1 million documents. Which model would you use for search: bi-encoder or cross-encoder? Why?

❓ Question

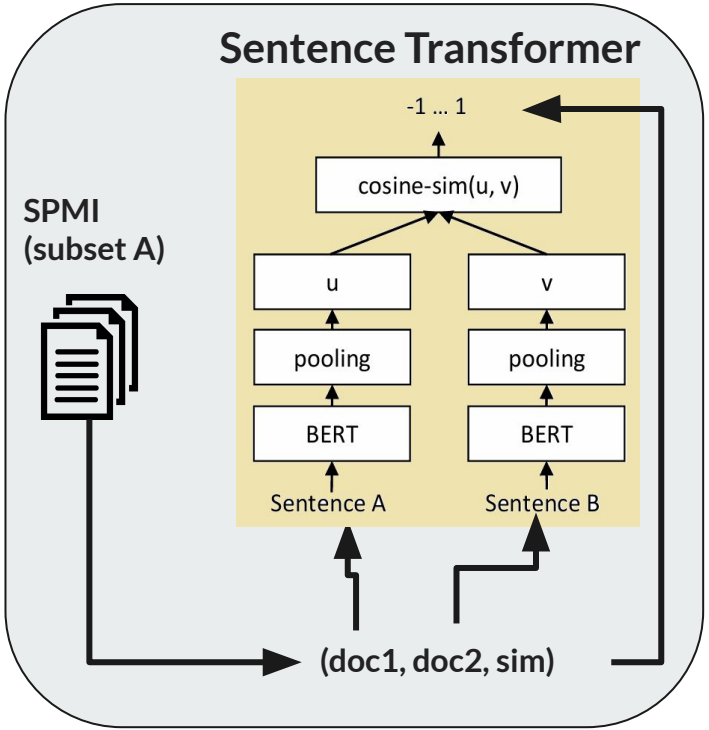
> When would you still prefer a cross-encoder despite its cost?

❓ Question

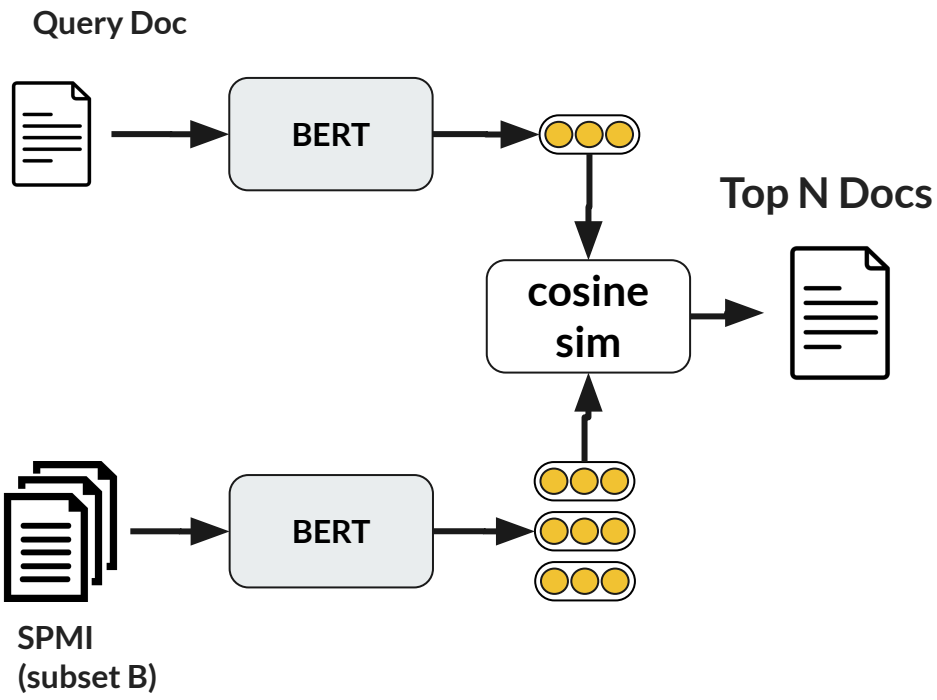
> Which is more important for a search engine: speed or accuracy? Can you have both?

RoadMap

Model Training



Model Inference



Toolkits

```
#Calculo de combinações
from itertools import combinations
combinations(lista, 2) # pares

#Balancear dados
from sklearn.utils import resample
undersampled_majority_class = resample(majority_class,
                                       replace=False,     # Don't duplicate samples
                                       n_samples= len(minority_class),  # Match minority
                                       random_state=42)

#Criar train, test split estratificados
from sklearn.model_selection import train_test_split
scores = [score for _, _, score, *_ in balanced_abstract_pairs]
train_data, test_data = train_test_split(
    balanced_abstract_pairs,
    test_size=0.2,
    random_state=42,
    stratify=scores
)
```

Model training

```python
from sentence_transformers import SentenceTransformerTrainer, SentenceTransformerTrainingArguments
from sentence_transformers.similarity_functions import SimilarityFunction
from sentence_transformers.evaluation import EmbeddingSimilarityEvaluator
```

```python
args = SentenceTransformerTrainingArguments(
    # Required parameter:
    output_dir="models/sentence_transformers/bert-base-portuguese",
    report_to="none",
    # Optional training parameters:
    num_train_epochs=5,
    per_device_train_batch_size=16,
    per_device_eval_batch_size=16,
    fp16=True,  # Set to False if you get an error that your GPU can't run on FP16
    bf16=False,  # Set to True if you have a GPU that supports BF16
    # Optional tracking/debugging parameters:
    eval_strategy="epoch",
    save_strategy="epoch",
    save_total_limit=2,
    load_best_model_at_end=True,
)
```

Model training

```
# Create the evaluator
dev_evaluator = EmbeddingSimilarityEvaluator(
    test_dataset['abstract1'],  # Assuming these are the sentence pairs for evaluation
    test_dataset['abstract2'],
    test_dataset['score'],  # Assuming this contains the similarity scores
    main_similarity=SimilarityFunction.COSINE,
)

# 6. Create the trainer & start training
trainer = SentenceTransformerTrainer(
    model=model,
    args=args,
    train_dataset=train_dataset,
    eval_dataset=test_dataset,
    loss=loss,
    evaluator=dev_evaluator,
)
```

Inference

```
from sentence_transformers import util
import torch

embeddings = model.encode(abstracts, convert_to_tensor=True)
query_embedding = model.encode(query_text, convert_to_tensor=True)

# Calculate the similarity between the query and the abstracts
cosine_scores = util.pytorch_cos_sim(query_embedding, embeddings)
retrieval_results = torch.topk(cosine_scores, k=15)

retrieval_results.values  -> lista de pares (score, index)
```

S/
Identidicação: género feminino, 24 anos.

AP:
sem antecedentes conhecidos.

MH:
contraceptivo oral

HDA: Recorre ao serviço de urgência por **nódulos violáceos dolorosos na região pré-tibial bilateral,** que estenderam a toda a perna e coxa, com 1 mês de evolução. Três semanas antes do início do quadro, realizou a 2ª dose da vacina contra **SARS-CoV-2** Comirnaty® (Pfizer-BioNTech), a mesma que realizou um mês antes. Sem outras queixas, nomeadamente sugestivas de patologia auto-imune ou síndrome onstitucional.

O/
Exame físico- apirética, sem alterações da orofaringe ou adenopatias, observando-se lesões nodulares em ambas as coxas e pernas, de cor violácea, eritematosas, com 2 cm de maior eixo, dolorosas à palpação.

A/
Analiticamente- **leucocitose de 12,6 x 109/L com neutrofilia, proteína C reactiva 210 mg/L e velocidade de sedimentação 42 mm/h.**
Anticorpo anti-estreptolisina O ne-gativo, serologias para sífilis e vírus Ebstein-Barr, hepatite B e C e VIH negativas. Painel imunológico, com pesquisa de anticorpos anti-nucleares (ANA), anticorpos anti-double--stranded DNA (dsDNA), anticorpos itoplasmáticos anti--neutrófilos (ANCA), complemento, factor reumatóide e anticorpos anti-péptido citrulinado cíclico, negativos.
Marcadores de lesão hepática e função renal sem alterações.
Hemoculturas negativas.
TC TAP sem alterações.

Assim, assumido **eritema nodoso secundário à administração da vacina contra SARS-CoV-2.**
Iniciada **prednisolona 60** mg/dia.

Evolução:
Uma semana depois houve melhoria clínica significativa e diminuição acentuada dos parâmetros inflamatórios, pelo que se iniciou redução gradual de corticoterapia, com resolução completa do quadro ao fim de duas semanas.
Passados seis meses, a doente teve doença ligeira por SARS-CoV-2 e uma semana depois surgiram lesões com características idênticas às do episódio anterior. Iniciou novo ciclo de corticoterapia, com resolução do quadro. Pela exuberância das lesões, a doente foi aconselhada a não repetir imunização com vacina de mRNA contra SARS-CoV-2.
Até à data, não voltou ter infecção por SARS-CoV-2 nem teve ressurgimento das lesões cutâneas.

S/
Identificação: Género feminino, 31 anos

AP:
Puérpera IGIP
parto eutócico às 40 semanas e 4 dias de gestação
História de 3 episódios de bacteriúria assintomática, com uroculturas positivas para Escherichia coli durante a gravidez tratadas com fosfomicina.
fez suplementação oral com ácido fólico e ferro durante a gravidez. Restantes antecedentes pessoais sem relevo no presente contexto.

HDA: Ao 3º dia de pós-parto foi realizada visita de Ginecologia para decisão de alta clínica. À avaliação clínica por Ginecologia e Obstrícia- "hemodinamicamente estável e apirética, **pálida com escleróticas ligeiramente ictéricas**. Lóquios normais, mamas tensas e períneo com cicatriz de episiorrafia sem sinais inflamatórios. Foram realizadas análises de controlo que revelaram **anemia normocítica normocrómica e trombocitopénia grave**"
Neste contexto, pedida avaliação urgente por Medicina Interna.
À minha avaliação, refere cansaço e cefaleia holocraniana que atribui ao pós-parto e à privação de sono. Nega hematúria, rectorragia, gengivorragia ou outras perdas hemorrágicas.

O/
Exame físico sem hematomas, petéquias ou equimoses. Sem outras manifestações neurológicas.

A/
Analiticamente com sinais evidentes de a**nemia hemolítica com elevação da LDH, haptoglobina indoseável e elevação da bilirrubina à custa da indireta, com Coombs direto e indireto negativos**. Sem consumo de complemento.
esfregaço de sangue periférico com **anisocitose, policromatofilia, 1,2% de esquizócitos, 1 eritroblasto por cada 100 leucocitos, raras plaquetas grande**s.
ecografia abdominal sem esplenomegália ou outras alterações.
serologias virais foram negativas, bem como a pesquisa de marcadores de autoimunidade. **# anticorpo anti-ADAMTS13 positivo fraco com um valor de 17 UI/mL (negativo < 13 UI/mL) e a atividade ADAMTS13 foi de 6% (positivo < 10%**).

Assim, assumida púrpura trombocitopénica trombótica
Iniciada plasmaférese cerca de 6 horas após o início da suspeita clínica e completou 5 sessões, em dias consecutivos. Realizou3 dias de metilprednisolona 1g/dia.
Com melhoria analítica progressiva e estabilização dos valores de hemograma e resolução dos sintomas neurológicos.
Até à atualidade não apresentou recidiva.

Information Retrieval

Sentence BERT

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